Storage Lab: End-to-End Cloud Block Volume Resize & XFS/EXT4 Filesystem Growth

Duration : 60 Min.

Scenario

A cloud virtual machine has its attached block volume /dev/nvme1n1 resized from 100 GB to 250 GB in the cloud control plane. However, the OS partition /dev/nvme1n1p1 and its XFS filesystem mounted at /var/lib/pgsql still register 100 GB. The application database is actively running and cannot experience downtime. The engineer must use growpart to expand partition 1, use xfs_growfs to extend the XFS filesystem online, verify zero loss of running database transactions, and document the full execution sequence.

Task Objectives

Perform the following zero-downtime cloud block volume expansion tasks:

> **Pre-Staged Files in `student_workspace/`:** - **`cloud_resize.log`**: Starter execution log with `TODO_*` placeholders.

1. Inspect the enlarged cloud block device `/dev/nvme1n1` and identify the unallocated disk partition space.
2. Expand the active OS partition to consume the full available volume capacity without corrupting the partition table.
3. Extend the live XFS filesystem online while database I/O operations continue in the background.
4. Verify that the expanded filesystem capacity is recognized and document execution steps in `cloud_resize.log`.

Instructions to Perform the Task

1. Open your assessment **Terminal** or storage management CLI interface.
2. Execute the required storage management commands, volume operations, or configuration edits.
3. If configuration files, backup manifests, or diagnostic reports are required, save them inside the `student_workspace` directory.
4. Verify your storage configurations locally by executing: `bash python run.py`
5. Once verified, return to the platform dashboard and click **"Run Tests"** / **"Full Eval"** to evaluate your score!

Validation

Once your storage tasks are executed, return to the platform dashboard and click the **"Run Tests"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Disk geometry inspection: correctly identify discrepancy between disk device size and partition size via lsblk	5 Marks
TC2	Partition table extension: growpart successfully expands partition 1 without corrupting partition GUID table (GPT)	5 Marks
TC3	Online filesystem expansion: xfs_growfs extends the live XFS filesystem online without requiring unmount	5 Marks
TC4	Integrity & verification: database writes verified intact and df -hT confirms ~250 GB available space	5 Marks

Total Score: 20 Marks

Important Notes

- This is an **Auto-Evaluated** storage assessment.
- Ensure all created volumes, files, and scripts are properly configured and saved before submitting.